

# CROSSCERT

Web-Based Event Management & Certificate Automation Platform. Built for Holy Cross of Davao College — Office of the Vice President for Academic Affairs.

📅 March 2026

🔄 Norman's Model · ACM Framework

👤 Admin + Participant Guide

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## 01 System Overview

Platform capabilities and feature summary

CROSSCERT is a next-generation, web-based platform that digitizes the full lifecycle of institutional events — from creation and registration, to attendance tracking and automated certificate delivery. Built for **Holy Cross of Davao College**, it serves two core user roles: Admins who manage events, and Participants who register and receive certificates.

| FEATURE           | WHO USES IT | WHAT IT DOES                                                   |
|-------------------|-------------|----------------------------------------------------------------|
| Event Management  | Admin       | Create, manage, and publish events with full status control    |
| Digital Tickets   | Participant | Receive encrypted QR access passes upon registration           |
| QR Attendance     | Admin       | Scan participant QR codes for real-time check-in and check-out |
| Evaluations       | Participant | Submit post-event feedback to trigger certificate release      |
| Certificates      | Both        | Auto-generate and deliver hi-res PDF certificates via email    |
| Insights          | Admin       | View attendance, engagement, and participation analytics       |
| Dark / Light Mode | Both        | Toggle theme for your preferred display environment            |

## Technology Stack

**Frontend** Next.js 15 · TypeScript · Tailwind CSS · Radix UI

**Database** PostgreSQL / SQLite

**Backend** Django 5.1 · Python · ReportLab PDF

**Messaging** Automated SMTP Email Delivery

02

## Interaction Frameworks

Norman's Model and the ACM HCI Framework

### Norman's 7-Stage Model of Interaction

Every feature in CROSSCERT is mapped to Donald Norman's seven-stage cycle, so users always know what to do, what is happening, and what to expect from the system at every point.

**1 Goal**  
What do I want to accomplish?

**2 Plan**  
What are my options?

**3 Specify**  
What specific action do I take?

**4 Perform**  
I click, type, or scan.

**5 Perceive**  
What does the system show me?

**6 Interpret**  
What does the feedback mean?

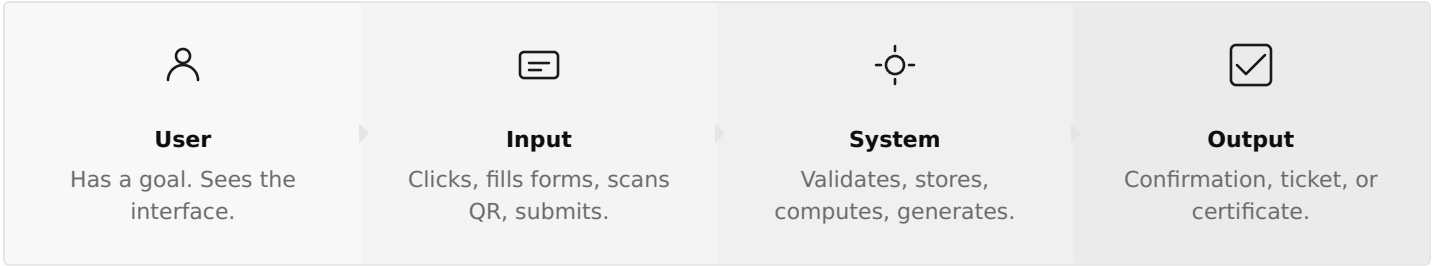
**7 Compare**  
Did the system do what I intended?

Applied Example — Registering for an Event

| STAGE       | USER EXPERIENCE IN CROSSCERT                           |
|-------------|--------------------------------------------------------|
| 1 Goal      | I want to attend the upcoming seminar.                 |
| 2 Plan      | Navigate to Events list, find the seminar.             |
| 3 Specify   | Click the event card to open its landing page.         |
| 4 Perform   | Click the <b>Register Now</b> button.                  |
| 5 Perceive  | A success modal appears; QR ticket delivered to email. |
| 6 Interpret | My registration is confirmed.                          |
| 7 Compare   | QR ticket is visible in My Events — goal achieved.     |

ACM Framework — User · Input · System · Output

The ACM HCI Framework models interaction as a continuous loop. Every action in CROSSCERT follows this four-stage cycle.



03

User Roles

Admin and Participant capabilities

✓ Admin

✓ Create & manage events

✓ Control event lifecycle

✓ Scan QR for attendance

✓ Review evaluations

✓ Generate certificates

✓ View analytics & insights

✓ Manage system settings

🧑 Participant

✓ Discover & browse events

✓ Register for events

✓ Receive digital QR ticket

✓ Submit post-event evaluation

✓ Download PDF certificate

✓ Manage bookmarks

✓ View personal settings

Three primary flows drive the CROSSCERT experience, documented using the ACM User-Input-System-Output model.

Flow A — Event Creation Lifecycle (Admin)

| STEP     | USER INPUT                 | SYSTEM PROCESS                         | OUTPUT                        |
|----------|----------------------------|----------------------------------------|-------------------------------|
| Create   | Click "Create Event"       | Validates fields, stores record        | Event saved as Draft          |
| Publish  | Change status to Scheduled | Updates state, notifies users          | Event visible to participants |
| Go Live  | Change status to Live      | Enables QR scanner module              | Check-in module active        |
| Conclude | Change status to Concluded | Closes registration; opens evaluations | Evaluation forms sent         |

Flow B — Participant Registration & Check-In

| STEP        | USER INPUT               | SYSTEM PROCESS                                             | OUTPUT                          |
|-------------|--------------------------|------------------------------------------------------------|---------------------------------|
| Discover    | Browse events list       | Fetches all Scheduled events                               | Event cards displayed           |
| Register    | Click "Register Now"     | Validates eligibility, slots; creates ticket; generates QR | Success modal; QR sent to email |
| Check-In    | Show QR to admin scanner | Decodes QR, logs timestamp                                 | Checked In — recorded           |
| After Event | —                        | Admin sets Concluded status                                | Evaluation prompt appears       |

Flow C — Evaluation & Certificate Issuance

| STEP     | USER INPUT                           | SYSTEM PROCESS                                         | OUTPUT                           |
|----------|--------------------------------------|--------------------------------------------------------|----------------------------------|
| Evaluate | Submit evaluation form               | Stores response; marks participant as evaluated        | Submission confirmed             |
| Generate | Admin clicks "Generate Certificates" | Checks eval status; runs ReportLab PDF engine          | Hi-res PDF built per participant |
| Deliver  | —                                    | SMTP sends PDF to each participant; saves to portfolio | Email sent + download available  |

**Important:** Certificates are not generated until the participant has submitted their evaluation. This is required to trigger the PDF generation process.

## 05 System Status Messages

All platform feedback states and their meaning

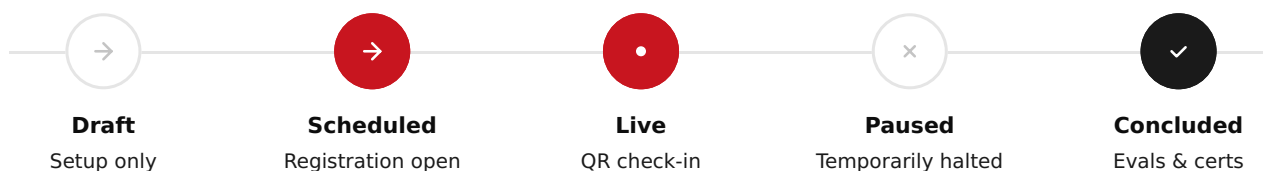
| STATUS                  | MEANING                         | WHAT TO DO                                      |
|-------------------------|---------------------------------|-------------------------------------------------|
| Registration Successful | You are registered              | Check your email for the QR ticket              |
| Check-In Recorded       | Attendance captured             | You may now enter the event                     |
| Evaluation Submitted    | Feedback saved                  | Wait for certificate generation                 |
| Certificate Ready       | PDF generated                   | Go to /participant/certificates to download     |
| Already Registered      | Duplicate registration detected | Check My Events for your existing ticket        |
| Event Full              | No slots remaining              | Bookmark for future announcements               |
| Invalid QR Code         | Code unrecognized               | Ensure you are showing the correct event ticket |
| Evaluation Required     | Certificate blocked             | Submit your evaluation first                    |

## 06 Event Lifecycle

State-driven event management

### Event Status Progression

Every event passes through a defined sequence of statuses. Each status controls what participants and admins can do.



| STATUS    | MEANING                         | PARTICIPANT CAN REGISTER? |
|-----------|---------------------------------|---------------------------|
| Draft     | Event is being set up           | No                        |
| Scheduled | Event is published and upcoming | Yes                       |
| Live      | Event is currently happening    | QR check-in active        |
| Paused    | Temporarily suspended           | No new check-ins          |
| Concluded | Event has ended                 | Evaluations open          |

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## UI Design & Visual Language

Color system, typography, animation, and component patterns

### Design Philosophy

| PRINCIPLE   | HOW IT MANIFESTS                                                   |
|-------------|--------------------------------------------------------------------|
| Clarity     | Minimal clutter — every element on screen has a purpose            |
| Consistency | Same color, spacing, and interaction patterns across all pages     |
| Feedback    | Every user action produces immediate, visible confirmation         |
| Delight     | Smooth animations and glassmorphism effects make the UI feel alive |

### Color System — Brand Primary: Institutional Red

**Light Mode**  
hsl(354, 86%, 40%) · #B80B1C

**Dark Mode**  
hsl(0, 100%, 50%) · #FF0000

Gradient form: from-red-600 to-rose-600

Department Color Coding — Glassmorphism Badges

● STE

● CET

● SBME

● CHATME

● HUSOCOM

● COME

● CCJE

All badges use `backdrop-blur-md` for the frosted glass effect throughout event cards.

Typography

# Syne

DISPLAY FONT · WEIGHT 400-800

Headings, hero text, display titles, sidebar navigation labels.

# Space Grotesk

BODY FONT · WEIGHT 300-700

Body text, labels, navigation, forms, and all readable content.

Animation System

| ANIMATION     | DESCRIPTION & USAGE                                                                                          |
|---------------|--------------------------------------------------------------------------------------------------------------|
| float         | Gentle vertical oscillation (0 to -20px). Used on hero visual elements.                                      |
| blob          | Organic scale and translate morphing. Used on background glow decorations.                                   |
| scan          | Top to bottom sweep with opacity fade. Used on QR scanner overlay.                                           |
| marqueeSlide  | Infinite left scroll. Used on the department badge ticker strip on the landing page.                         |
| Framer Motion | Splash entrance: scale 0.8 to 1, opacity 0 to 1. Exit: -y100 opacity 0 over 0.8s. Easing [0.76, 0, 0.24, 1]. |

**3D Lanyard:** The landing page features a physics-based 3D lanyard built with React Three Fiber and @react-three/rapier. The CROSSCERT ID card is draggable in real-time.

Iconography — Lucide React

| ICON            | USED FOR             | ICON      | USED FOR                 |
|-----------------|----------------------|-----------|--------------------------|
| LayoutDashboard | Dashboard navigation | Calendar  | Events section           |
| Users           | Participants section | QrCode    | QR Attendance / Check-In |
| FileText        | Certificates         | BarChart3 | Insights / Analytics     |
| Bookmark        | Bookmarks            | Award     | Participant Certificates |
| Sparkles        | Role badge indicator | LogOut    | Logout button            |





## Step 1 — Log In to the

### ✓ Platform

Path: crosscert.hcdc.edu.ph

#### 1 Navigate to the CROSSCERT URL

Open your browser and go to the platform URL. The animated splash screen and 3D lanyard loads on the landing page.

#### 2 Click "Sign In" and enter your credentials

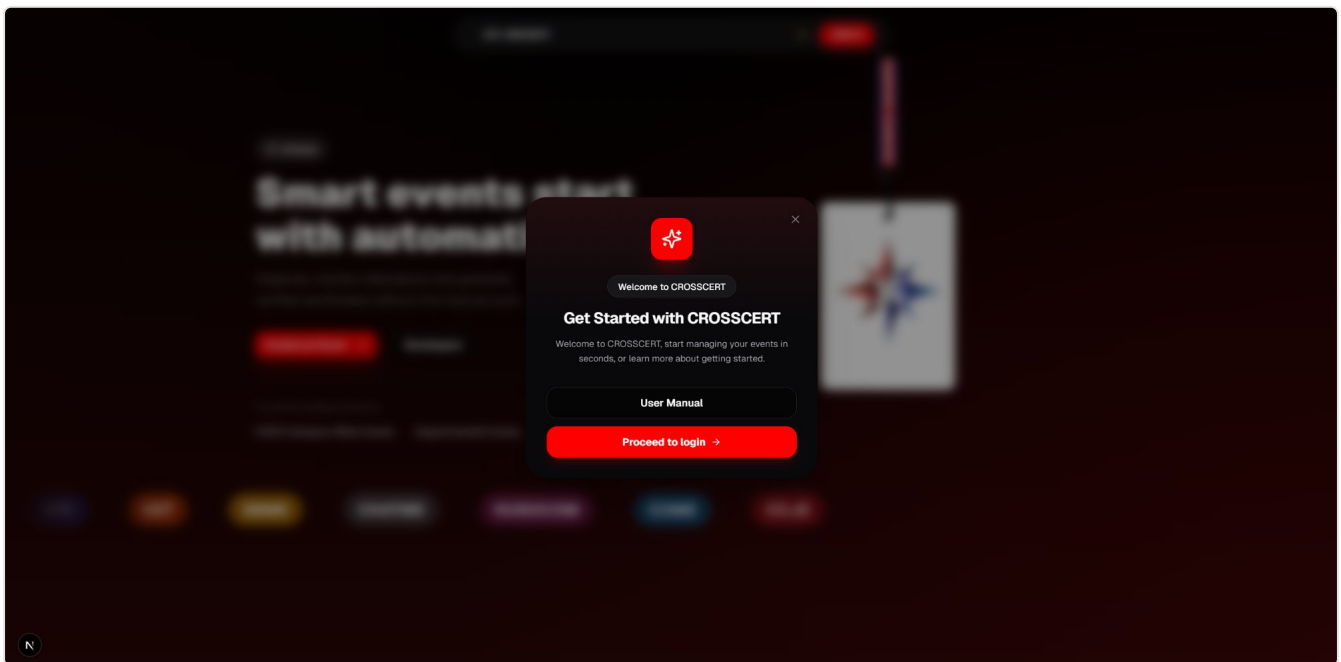
Enter your Email and Password, then click Sign In.

Output: You land on /participant/dashboard

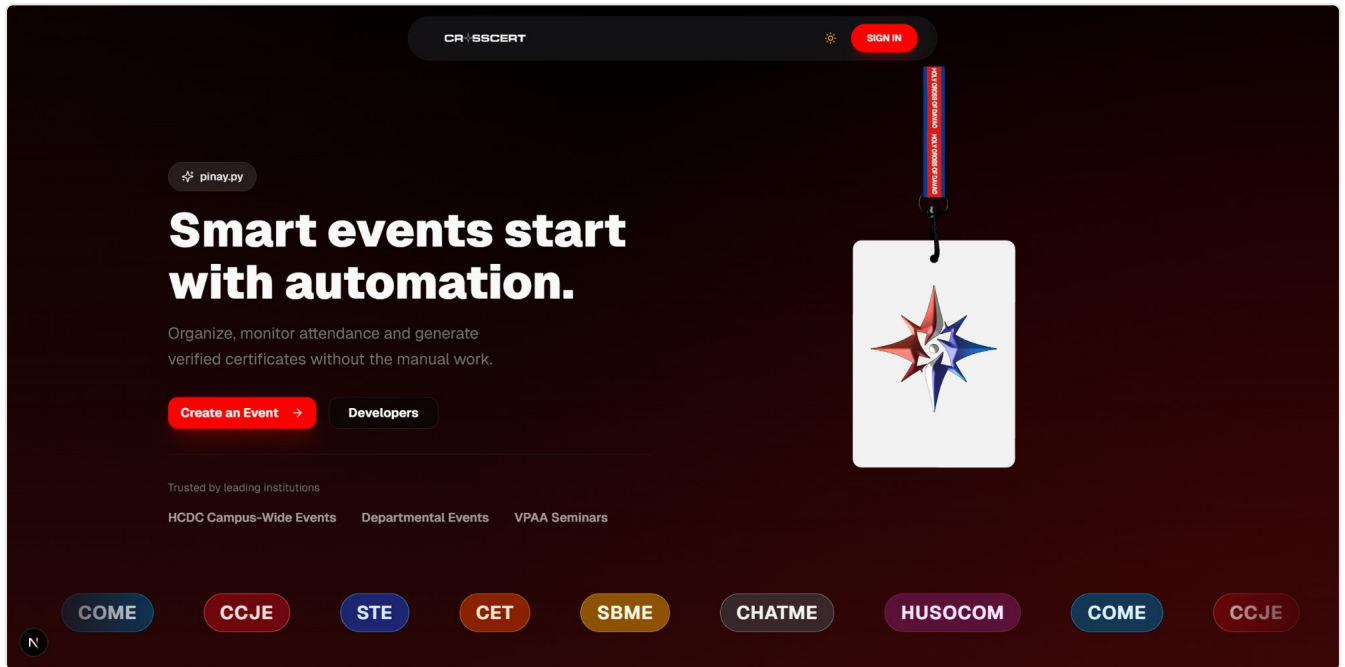
#### 3 No account? Sign Up first

Click Sign Up, fill in Full Name, Email, Department, Password, and Confirm Password. Verify your email before returning to Sign In.

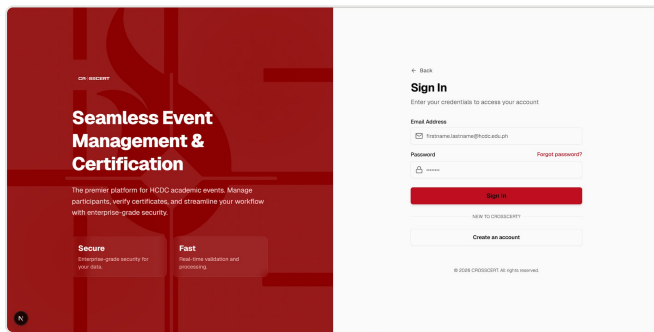
### — WELCOME MODAL — USER MANUAL OR PROCEED TO LOGIN



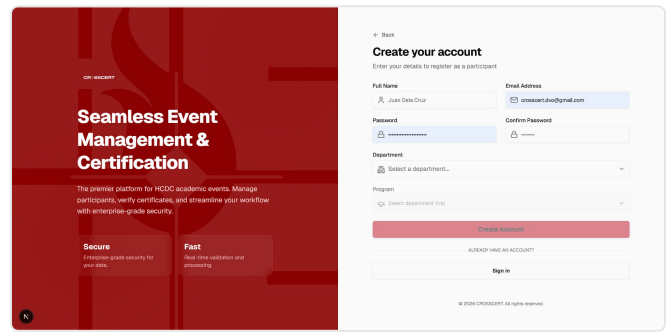
*First-time visitors see this modal — click "User Manual" to access this guide, or "Proceed to login" to go straight to the platform.*



— SIGN IN



— CREATE ACCOUNT

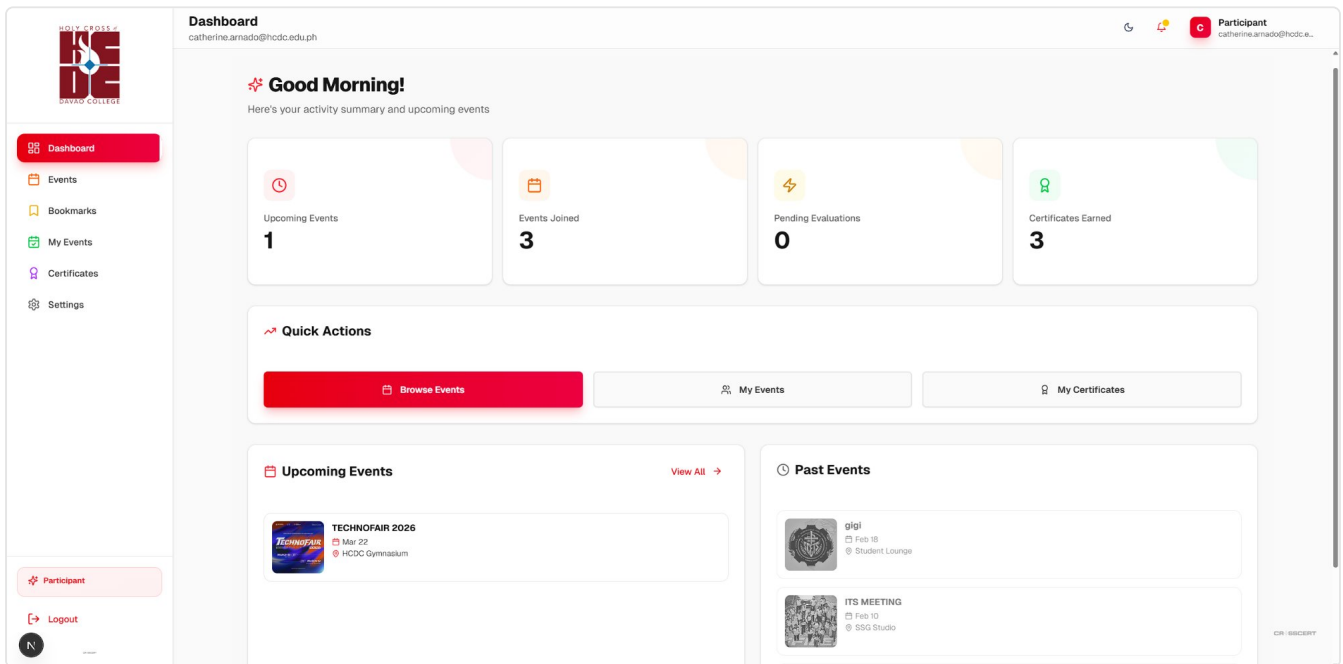


**1 Review Upcoming Events**

Events you are already registered for appear at the top with their current status.

**2 Check Notifications and Certificates**

New event alerts and certificate-ready messages from admins appear in the notification panel. Recently issued PDFs appear for quick access.

**— PARTICIPANT DASHBOARD**

The screenshot displays the Participant Dashboard for a user named Catherine Amado. The interface includes a sidebar with navigation links: Dashboard (selected), Events, Bookmarks, My Events, Certificates, and Settings. The main content area features a 'Good Morning!' greeting and a summary of activity. Key statistics shown are: 1 Upcoming Event, 3 Events Joined, 0 Pending Evaluations, and 3 Certificates Earned. Below these are 'Quick Actions' buttons for 'Browse Events', 'My Events', and 'My Certificates'. The 'Upcoming Events' section lists 'TECHNOFAIR 2026' on Mar 22 at HCCDC Gymnasium. The 'Past Events' section lists 'gigi' on Feb 18 at Student Lounge and 'ITS MEETING' on Feb 10 at SSG Studio. A 'Logout' button is visible in the bottom left corner.

**Dashboard**  
catherine.amado@hcdc.edu.ph

**Good Morning!**  
Here's your activity summary and upcoming events

**Upcoming Events** [View All →](#)

- TECHNOFAIR 2026**  
Mar 22  
HCCDC Gymnasium

**Past Events**

- gigi**  
Feb 18  
Student Lounge
- ITS MEETING**  
Feb 10  
SSG Studio

**Participant**  
[Logout](#)

## Step 3 — Discover & Browse

### Events

Path: /participant/events or /discover

#### 1 Click Events in the sidebar

A grid of event cards appears, each showing Event Name, Date and Time, Department badge, available slots, and a bookmark icon.

#### 2 Search or Filter events

Use the Search bar to find a specific event by name. Filter by department or date if needed.

#### 3 Open the Event Landing Page

Click any card to open the full event page: cover banner, description, date, location, slots remaining, and the Register button.

### — DISCOVER EVENTS

The screenshot displays the 'Discover Events' interface. On the left is a sidebar with navigation links: Dashboard, Events (highlighted), Bookmarks, My Events, Certificates, and Settings. Below these are buttons for 'Participant' and 'Logout'. The main content area is titled 'Events' with the user email 'catherine.amado@hcdc.edu.ph'. It features a 'Discover Events' section with a search bar and filters for School Year (All Years) and Semester (All Semesters, 1st Semester, 2nd Semester, Summer). A horizontal menu shows department filters: ALL, HDCD EVENTS, CCJE, CET, CHATME, HUSOCOM, COME, SBME, and STE. Below this, tabs for 'Upcoming (1)' and 'Past Events (4)' are visible. The main event card for 'TECHNOFAIR 2026' is highlighted, showing the date 'MARCH 18 (D1)' and location 'HDCD Gymnasium', with a 'Join Event' button. A 'FILTER BY MONTH' dropdown on the right lists months from JANUARY to DECEMBER, with 'All Months' selected. The bottom right corner includes a 'DR - SECURITY' label.

## Step 4 — Register for an

### Event

Core participant interaction

#### 1 Review the event details and click "Register Now"

Confirm the date, location, and description. Click the red Register Now button.

#### S System validates your registration

Checks: Are you signed in? Is the event Scheduled? Are there available slots? Are you not already registered? Then creates your registration record and generates an encrypted QR Digital Access Pass.

Output: Success modal + QR sent to your email

#### 2 Event appears in My Events

The registered event is now visible in your My Events section with real-time status tracking.

### — EVENT DETAILS & REGISTRATION

The screenshot displays the 'Technofair 2026' registration interface. On the left, a sidebar contains navigation links: Dashboard, Events, Bookmarks, My Events, Certificates, and Settings. The main content area features a large header for 'TECHNOFAIR 2026' with event details: Sun 22 Mar, 08:32 - 21:30, HCDC Gymnasium. Below this, there are filters for Semester (2nd Semester), School Year (2026-2027), Capacity (1000000), and Access (Free Entry). The 'Event Description' section states that the event is a multi-day event held at HCDC, bringing together students through games, competitions, and interactive activities. A 'Key Speakers' section lists Owen B Pilongo, Reynante Aliling, and Michelle Capitan. On the right, a 'COMPLETION' section shows a 'Verified Certificate' button. A floating modal on the right displays the user's registration status as 'REGISTERED' with a QR code and a 'View Ticket' button. The user's profile at the top right shows they are logged in as 'Participant' (catherine.arnado@hcdc.edu.ph).

**Event Full?** If no slots remain, the Register button is disabled. Bookmark the event to watch for announcements.

Step 5 — Use Your QR Ticket at the Event

Digital Access Pass — QR Check-In

- 1

**Find your QR ticket**

Check your email inbox (or spam folder) for the QR pass. Or go to My Events in the app and tap your event to view it directly.
- 2

**Present your QR at the entrance**

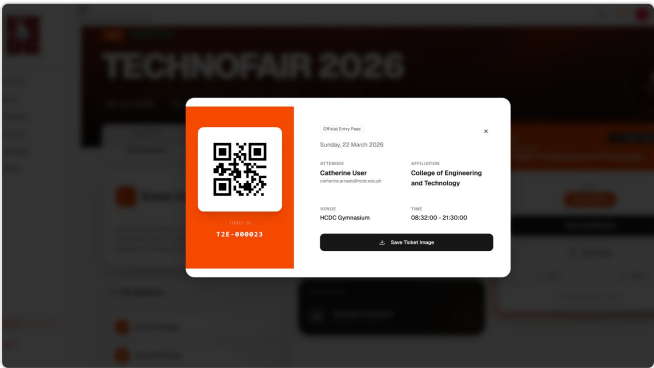
Maximize screen brightness. Hold the QR code flat and steady. Show it to the admin at the event entrance.
- S

**Admin scans with the CROSSCERT scanner**

The system decodes your QR, validates your registration, and logs the timestamp.

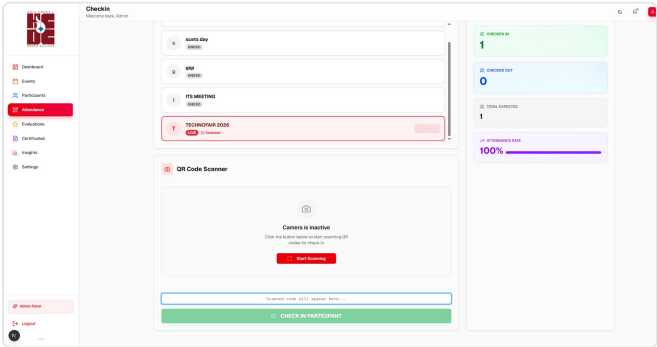
Output: Check-In Successful — attendance recorded

DIGITAL ACCESS PASS (QR TICKET)



Official Entry Pass — scannable at entrance

ADMIN QR CHECK-IN SCANNER



Admin scans participant QR to record attendance

## Step 6 — Submit Your

### ≡ Evaluation

Required before certificate issuance

#### 1 Go to My Events after the event concludes

Find the concluded event. An Evaluate button appears next to it. The event progress tracker shows REG → IN → OUT → EVAL stages.

#### 2 Complete the feedback form

Select a star rating (1–5) across multiple categories: Content Quality, Instructor/Resource Speaker, Facilities/Venue, and Overall Experience. Optionally write comments.

#### 3 Click "Submit Evaluation"

Your response is saved and your account is marked as evaluated, unlocking certificate generation.

Output: Thank you — evaluation confirmed

### — MY EVENTS — PROGRESS TRACKER (REG · IN · OUT · EVAL)



- Dashboard
- Events
- Bookmarks
- My Events**
- Certificates
- Settings

#### My events

catherine.arnado@hcdc.edu.ph

← Back

### My Events

Track your registrations, attendance, and feedback

Running & Upcoming (1)History (3)



EVALUATED

#### TECHNOFAIR 2026

🕒 08:32:00 📍 HCDC Gymnasium

✓ REG

—

✓ IN

—

✓ OUT

—

✓ EVAL

View Ticket

✓ Mark as Done

### — EVALUATION REMINDER MODAL



#### Pending Evaluations

Your feedback helps us improve!

You have 1 event(s) waiting for your feedback!

TECHNOFAIR 2026

March 22, 2026

Evaluate Now

Remind Me Later

View My Events

### — EVALUATION FORM

Back

#### Event Evaluation

Help us improve by sharing your experience

TECHNOFAIR 2026

Your Label

Rate Your Experience

Content Quality

★★★★☆ 4/5

Instructor/Resource Speaker

★★★★☆ 4/5

Facilities/Venue

★★★★☆ 4/5

Overall Experience

★★★★☆ 4/5



1 Wait for admin to generate certificates

After all evaluations are collected, the admin triggers PDF generation. You receive an email notification when ready — including Registration Confirmation, Attendance Verified, Evaluate Reminder, and Certificate emails.

2 Go to Certificates in the sidebar

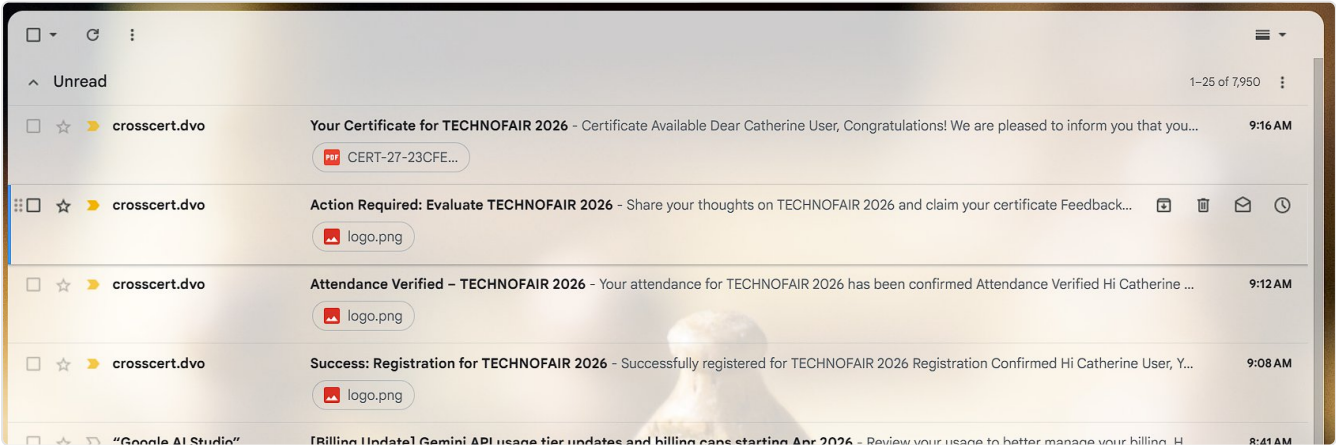
All your issued certificates appear as downloadable cards organized by event and date under "Access Granted."

3 Click "Download PDF"

A high-resolution, print-ready certificate opens or saves. It contains your Full Name, Event Title, Date, HCDC Institutional Header and Seal, Authorized Signatories, and a Unique Certificate ID.

Output: Print-ready PDF downloaded

— EMAIL NOTIFICATIONS — FULL FLOW



Registration Confirmation · Attendance Verified · Evaluate Reminder · Certificate Ready



Dashboard

Events

Bookmarks

My Events

Certificates

Settings

## Certificates

catherine.arnado@hcdc.edu.ph

← Back

# Access Granted

View and download your earned certificates and credentials



#CERT-27-...

### TECHNOFAIR 2026

ISSUED ON MARCH 19, 2026

This certifies that the recipient has successfully completed the requirements for the above event.

PDF



### ITS MEETING

ISSUED ON FEBRU

This certifies that completed the rec

— CERTIFICATE OF PARTICIPATION — PDF OUTPUT



Auto-generated hi-res PDF certificate with HCDC branding, CROSSCERT seal, and authorized signatories

## Admin Step 1 — Log In as

### ☒ Admin

Role-based login and dashboard access

#### 1 Navigate to CROSSCERT and click Sign In

Enter your Admin Email and Password, then click Sign In.

Output: /admin/dashboard with the Admin Sidebar

#### 2 Admin Sidebar Navigation

Access: Dashboard, Events, Participants, Attendance (QR), Evaluations, Certificates, Insights, and Settings.

## Admin Step 2 — View Admin

### Dashboard

Path: /admin/dashboard

#### 1 Review summary cards

At a glance: Active Events, Registered Participants, Checked In Today, Certificates Issued, and Evaluations Submitted.

#### 2 Upcoming events and real-time check-in activity

Upcoming events with their current statuses are listed below summary cards. Check-in activity updates in real-time.

## Admin Step 3 — Create an

### Event

Path: /admin/events

#### 1 Click Events in the sidebar, then "+ Create Event"

The event creation form opens.

#### 2 Fill in all required fields

Event Name · Description · Department · Date and Time · Location · Cover Image · Theme Color · Maximum Slots.

#### S Click "Create Event" — system validates and stores

Output: Event saved as Draft — not visible to participants yet

## Admin Step 4 — Manage Event

### Status

Advance through the lifecycle

#### 1 Open the event and find the Status control

Go to the Events list, click your event, then use the Status dropdown or button.

#### 2 Advance through statuses as needed

Draft → Scheduled: event publishes and registration opens. Scheduled → Live: QR scanner activates. Live → Concluded: evaluation forms sent to all attendees.

Status changes are instant — visible to participants live

## Admin Step 5 — Edit or Delete an

### Event

Path: /admin/events

#### 1 To Edit: click the pencil icon on the event card

Modify any field, then click Save Changes.

Output: Toast — Event updated successfully

#### 2 To Delete: click the trash icon and confirm

A confirmation dialog appears. Click Confirm Delete. The event and all associated data are permanently removed. This action cannot be undone.

## Admin Step 6 — Run QR Scanner (Check-

### In)

Path: /admin/checkin · Event must be Live

#### 1 Click Attendance, select the Live event, click "Start Scanner"

Your camera activates. Point it at the participant's QR ticket.

#### 5 System decodes QR and logs attendance

Success: Check-In Recorded — timestamp logged automatically

Error: Already Checked In / Invalid QR / Wrong Event

#### 2 Check-Out at end of event

Scan the same QR as participant exits. The system records the check-out timestamp and calculates attendance duration. A participant cannot check out before checking in.



## Admin Step 7 — Review Evaluations

Path: /admin/evaluations · Event must be Concluded

1

### Click Evaluations and select the event

All submitted evaluations appear with star ratings (1-5), written comments, participant name, department, and submission timestamp.

2

### Review and use for improvement

Use ratings and comments to assess event quality and inform future event planning.

**Note:** Only participants who submitted an evaluation will receive a certificate.

## Admin Step 8 — Generate



### Certificates

Path: /admin/certificates

1

### Click Certificates, select the Concluded event

The eligible participant list appears. Participants with a submitted evaluation are marked Ready. Those without are marked Blocked.

2

### Click "Generate Certificates"

ReportLab builds one certificate per eligible participant, applying event branding, name, date, and HCDC seal.

5

### System delivers and stores all certificates

Each PDF is emailed via SMTP to the participant's inbox and saved to their digital portfolio.

Output: Certificates issued to X participants

## Admin Step 9 — Configure



### Settings

Path: /admin/settings

| SECTION                | WHAT YOU CAN CONFIGURE                                              |
|------------------------|---------------------------------------------------------------------|
| Theme                  | Toggle Light / Dark mode platform-wide                              |
| Account Security       | Change admin password                                               |
| Email Configuration    | Set SMTP host, port, sender address for QR tickets and certificates |
| Institutional Branding | Upload updated HCDC logos and adjust platform color scheme          |

# 10 Quick Reference Map

Complete feature map for both roles

| ADMIN — FULL JOURNEY                                   | PARTICIPANT — FULL JOURNEY                          |
|--------------------------------------------------------|-----------------------------------------------------|
| 01 Log In as Admin — /admin/dashboard                  | 01 Sign Up or Log In — /participant/dashboard       |
| 02 View Admin Dashboard — summary cards and events     | 02 View Dashboard — events, notifications, certs    |
| 03 Create a New Event — fill form, save as Draft       | 03 Browse Events — /participant/events or /discover |
| 04 Publish Event — change status to Scheduled          | 04 Register for Event — click Register Now          |
| 05 Go Live — activate QR scanner module                | 05 Get QR Ticket — via email or My Events           |
| 06 Edit or Delete Event — pencil or trash icon         | 06 Show QR at Entrance — admin scans the code       |
| 07 Manage Participants — search, filter, view profiles | 07 Check My Events — track attendance and status    |
| 08 Run QR Scanner — check-in and check-out             | 08 Bookmark Events — save events for later          |
| 09 Review Evaluations — after Concluded status         | 09 Submit Evaluation — after event concludes        |
| 10 Generate Certificates — for evaluated participants  | 10 Download Certificate — /participant/certificates |
| 11 View Insights — /admin/insights analytics           | 11 Manage Settings — profile, password, theme       |
| 12 Configure Settings — SMTP, branding, password       | 12 Log Out — bottom of the sidebar                  |

CRSSCERT

EXCELLENCE IN MOTION

INSTITUTION

Holy Cross of Davao  
College

Office of the VPAA  
Davao City,  
Philippines

FRAMEWORK

Norman's Model  
(1988)

ACM HCI Framework  
Version 1.0 · March  
2026

DEVELOPMENT  
TEAM

Arnado, Catherine

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Supervised by John Rey Silverio · Intended for students, faculty, and administrators of Holy Cross of Davao College ·

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